ECE 310 Fall 2026
Lecture 16: Magnitude and Phase Response

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Learning Objectives

After this lecture, you should be able to:

- Compute and plot the magnitude response of an LTI system.
- Compute and plot the phase response of an LTI system.
- Define and explain the significance of group delay using the phase response.



Complex Numbers Review

Recall that we may write complex numbers $z \in \mathbb{C}$ in rectangular form

$$z = a + jb, \; z^* = a - jb$$

and polar/exponential form

$$z = Re^{j\theta}, \; z^* = Re^{-j\theta}.$$

We compute the *magnitude* and *phase* of complex numbers as follows.

$$\text{Magnitude}(z) = |z| = R = |\sqrt{zz^*}| = \sqrt{a^2 + b^2}$$

$$\text{Phase}(z) = \angle z = \theta = \begin{cases} \tan^{-1}\left(\frac{\text{Im}\{z\}}{\text{Re}\{z\}}\right), & a > 0 \\ \tan^{-1}\left(\frac{\text{Im}\{z\}}{\text{Re}\{z\}}\right) + \pi, & a < 0, \; b > 0 \\ \tan^{-1}\left(\frac{\text{Im}\{z\}}{\text{Re}\{z\}}\right) - \pi, & a < 0, \; b < 0 \end{cases}$$



Magnitude and Phase Response

Magnitude Response

The *magnitude response* of an LTI system is the magnitude of the system's frequency response for each $\omega \in [-\pi, \pi]$:

$$|H_d(\omega)| = \left|\sqrt{H_d(\omega)H_d^*(\omega)}\right|$$

- Remember that magnitude is always non-negative! $(|H_d(\omega)| \geq 0)$

Phase Response

The *phase response* of an LTI system is the phase of the system's frequency response for each $\omega \in [-\pi, \pi]$.

$$\angle H_d(\omega) = \tan^{-1}\left(\frac{\mathrm{Im}\{H_d(\omega)\}}{\mathrm{Re}\{H_d(\omega)\}}\right)$$

- Changes in sign for $\mathrm{Re}\{H_d(\omega)\}$ let to $\pm\pi$ jumps in phase, $-1 = e^{\pm j\pi}$!



Example 1: Plotting Magnitude and Phase Response

The impulse response of an LTI system is given by $h[n] = \{\underset{\uparrow}{1}, 0, 1\}$.

(a) Determine $H_d(\omega)$, $|H_d(\omega)|$, and $\angle H_d(\omega)$.

(b) Plot $|H_d(\omega)|$ and $\angle H_d(\omega)$.

(a) $H_d(\omega) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} = 1 \cdot e^{-j0} + 1 \cdot e^{-j2\omega} = 1 + e^{-j2\omega}$

$= e^{-j\omega}\left(e^{j\omega} + e^{-j\omega}\right)$

$= e^{-j\omega}(2\cos(\omega))$

$\rightarrow |H_d(\omega)| = \left|\sqrt{(2e^{j\omega}\cos(\omega))(2e^{-j\omega}\cos(\omega))}\right|$

$= \left|\sqrt{4\cos^2(\omega)}\right|$

$= 2|\cos(\omega)|$

Concept Check: π Jumps in Phase

Consider the following frequency responses of LTI systems. Which of the following systems will have $\pm\pi$ jump discontinuities in their phase response? (Select all that apply)

1. $H_1(\omega) = \cos(4\omega)$
2. $H_2(\omega) = e^{-j2\omega}(2 + \cos(2\omega))$
3. $H_3(\omega) = je^{-j3\omega}(\sin(3\omega))$
4. $H_4(\omega) = e^{-j2\omega}(1 - 2\cos(\omega) - \cos(2\omega))$
5. $H_5(\omega) = e^{-j2\omega}(3 - 2\cos(\omega) - \cos(2\omega))$



Example 2: Plotting Magnitude and Phase Response

The impulse response of an LTI system is given by $h[n] = \{\underset{\uparrow}{1}, 0, -1\}$.

(a) Determine $H_d(\omega), |H_d(\omega)|$, and $\angle H_d(\omega)$.

(b) Plot $|H_d(\omega)|$ and $\angle H_d(\omega)$.



Example 3: Plotting Magnitude and Phase Response

The impulse response of an LTI system is given by $h[n] = \{\underset{\uparrow}{3}, 2, 1\}$.

(a) Determine $H_d(\omega), |H_d(\omega)|$, and $\angle H_d(\omega)$.

(b) Plot $|H_d(\omega)|$ and $\angle H_d(\omega)$.

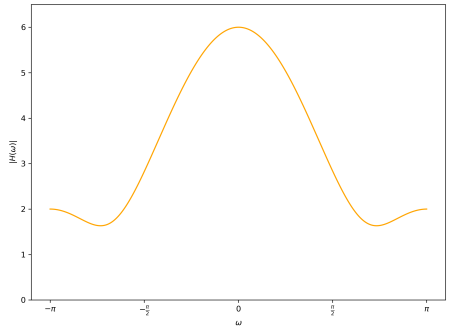


Example 3: Plotting Magnitude and Phase Response

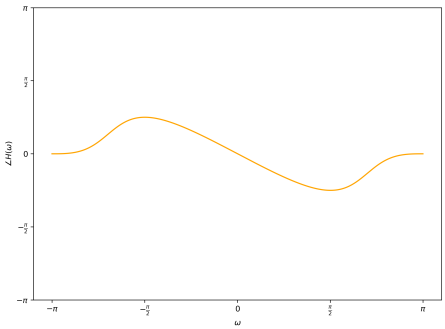
The impulse response of an LTI system is given by $h[n] = \{3, \underset{\uparrow}{2}, 1\}$.

(a) Determine $H_d(\omega), |H_d(\omega)|$, and $\angle H_d(\omega)$.

(b) Plot $|H_d(\omega)|$ and $\angle H_d(\omega)$.



(a) $|H_d(\omega)|$



(b) $\angle H_d(\omega)$

What does the phase response do?

Consider the frequency response $H_d(\omega) = e^{-j2\omega}$. Thus, $|H_d(\omega)| = 1$, $\angle H_d(\omega) = -2\omega$.

- What is the response to $x[n] = e^{j\frac{\pi}{2}n}$?
- What is the response to $x[n] = e^{j\frac{3\pi}{4}n}$?
- What is the response to $x[n] = e^{j\omega_0 n}$?



What does the phase response do?

Consider frequency response $H_d(\omega) = e^{-jk\omega}$, $k \in \mathbb{Z}$: $|H_d(\omega)| = 1$, $\angle H_d(\omega) = -k\omega$.

- What is the response to $x[n] = e^{j\omega_0 n}$?



Group Delay

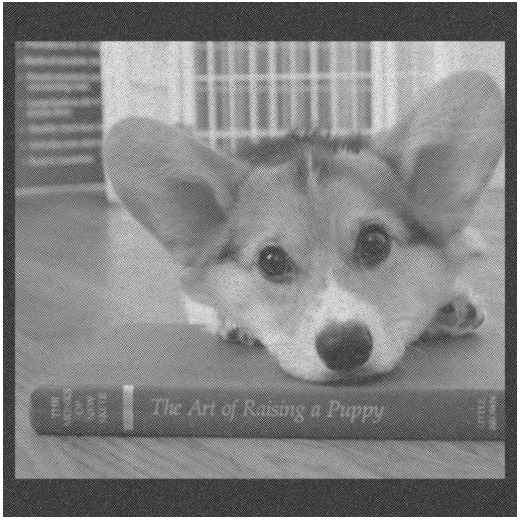
- The *group delay* τ_{gd} of a phase response is given by

$$\tau_{\mathrm{gd}}(\omega) = -\frac{d\angle H_d(\omega)}{d\omega}$$

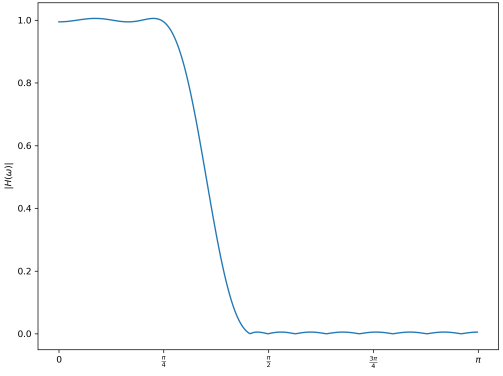
  and expresses how an LTI system will delay each frequency component in a signal.
- LTI systems with linear phase responses have *uniform group delay* and delay all frequencies by the same amount.
  - These systems have phase responses of the form $\angle H_d(\omega) = k\omega$.
  - For example, the previous example two slides ago.
  - The derivative of the phase response should be constant!
- LTI systems with non-uniform group delay shift different frequency components by different amounts
  - These systems have a phase responses with a non-constant derivative.



Group Delay Example

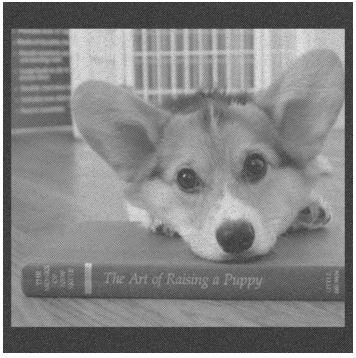


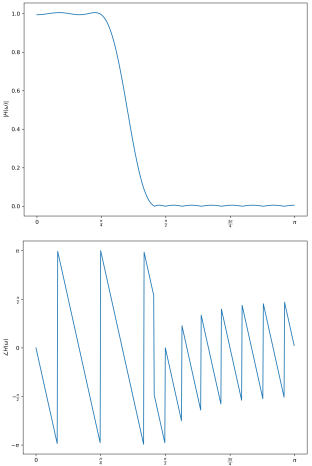
(a) Noisy image



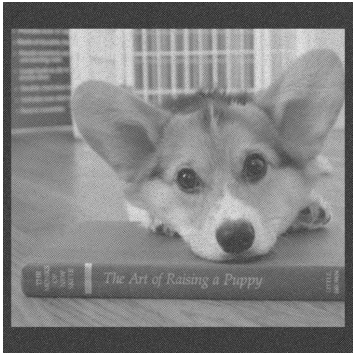
(b) Magnitude response of filter

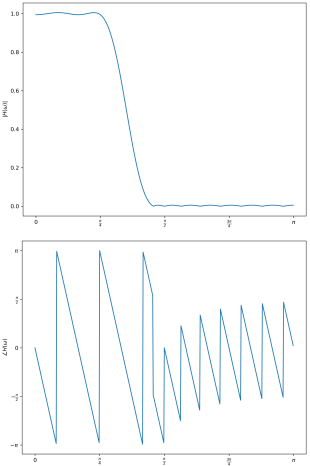
Group Delay Example

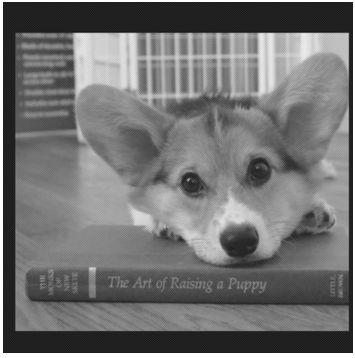




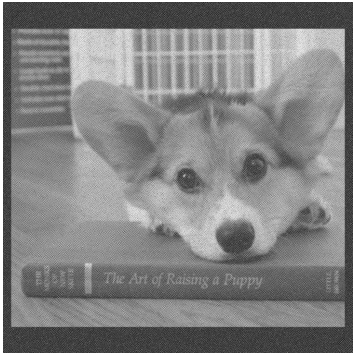
Group Delay Example

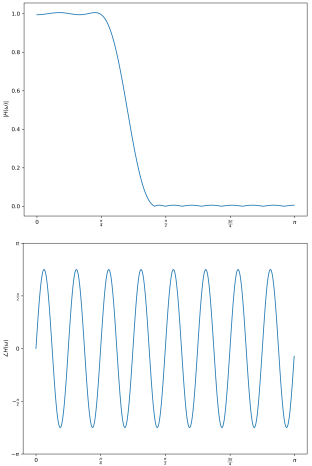






Group Delay Example





Group Delay Example

